



复数及其四则运算

Complex Numbers and Their Four Fundamental Operations

选择题 (Multiple-Choice Questions)

1. 下面 4 种说法中正确的是()

Which of the following 4 statements is correct?

- A. 如果实数 $a = b$, 则 $a - b + (a + b)i$ 是纯虚数; (If real numbers $a = b$, then $a - b + (a + b)i$ is a pure imaginary number;)
- B. 模相等的两个复数是共轭复数; (Two complex numbers with equal moduli are conjugate complex numbers;)
- C. 如果 z 是纯虚数, 则 $z \neq \bar{z}$; (If z is a pure imaginary number, then $z \neq \bar{z}$;)
- D. 任何数的偶次幂不小于 0. (The even power of any number is not less than 0.)

2. 下面给出 4 个不等式, 其中正确的是()

The following 4 inequalities are given, which one is correct?

- A. $3i > 2i$
- B. $|2 + 3i| > |1 - 4i|$
- C. $|2 - i| > 2i^4$
- D. $i^2 > -i$

3. $a = 0$ 是复数 $a + bi$ ($a, b \in R$) 为纯虚数的()

$a = 0$ is a(n) () for the complex number $a + bi$ (where $a, b \in R$) to be a pure imaginary number.

- A. 充分不必要条件 (Sufficient but not necessary condition)
- B. 必要不充分条件 (Necessary but not sufficient condition)
- C. 充分必要条件 (Sufficient and necessary condition)
- D. 既不充分又不必要条件 (Neither sufficient nor necessary condition)

4. 复数 $z = \sin \frac{\pi}{3} + \cos \frac{\pi}{6} i$, 则 $|z|^2 =$ ()

For the complex number $z = \sin \frac{\pi}{3} + \cos \frac{\pi}{6} i$, then $|z|^2 =$ ()

- A. $\frac{3}{4}$
- B. $\frac{\sqrt{6}}{2}$
- C. $\frac{3}{2}$
- D. 1

5. 已知复数 z 满足 $|z|^2 - 2|z| - 3 = 0$, 则复数 z 的对应点的轨迹方程是()

Given that the complex number z satisfies $|z|^2 - 2|z| - 3 = 0$, the trajectory equation of the corresponding point of the complex number z is ()

- A. 1 个圆 (One circle)
- B. 线段 (A line segment)
- C. 2 个点 (Two points)
- D. 2 个圆 (Two circles)

6. 方程 $z^2 - 5|z| + 6 = 0$ 的复数集内解的个数是()

The number of solutions of the equation $z^2 - 5|z| + 6 = 0$ in the set of complex numbers is ()

- A. 4
- B. 6
- C. 8
- D. 2

7. 若虚数 $z = a + bi$ ($a, b \in R$) 则 $|z^2|$ 、 $|z|^2$ 、 z^2 的关系是()

If the imaginary number $z = a + bi$ (where $a, b \in R$), then the relationship between $|z^2|$, $|z|^2$, and z^2 is ()

- A. 互不相等 (Mutually unequal)
- B. $|z^2| = |z|^2 \neq z^2$
- C. $|z^2| \neq |z|^2 = z^2$
- D. $|z^2| = |z|^2 = z^2$

8. 设 $z = (2t^2 + 5t - 3) + (t^2 + 2t + 2)i$ ($t \in R$), 则下列结论正确的是()

Let $z = (2t^2 + 5t - 3) + (t^2 + 2t + 2)i$ (where $t \in R$), then which of the following conclusions is correct?

- A. z 对应的点在第一象限 (The corresponding point of z is in the first quadrant)
- B. z 一定不是纯虚数 (z is definitely not a pure imaginary number)
- C. z 对应的点在实轴下方 (The corresponding point of z is below the real axis)
- D. z 一定不是实数 (z is definitely not a real number)

9. a, b 都是复数, 下列命题中, 错误的是()

Both a and b are complex numbers. Which of the following propositions is incorrect?

- A. 若 $a = b$, 则 $\bar{a} = \bar{b}$ (If $a = b$, then $\bar{a} = \bar{b}$)
- B. $\sqrt{|a|^2} = |a|$
- C. 若 $|a| = |b|$, 则 $a = \pm b$ (If $|a| = |b|$, then $a = \pm b$)
- D. 如果 $a = \bar{b}$, 则 $a + b$ 为实数 (If $a = \bar{b}$, then $a + b$ is a real number)

10. 若复数 $(a + 1) + (a - 1)i \neq 0$, 则 a 的值是()

If the complex number $(a + 1) + (a - 1)i \neq 0$, then the value of a is ()

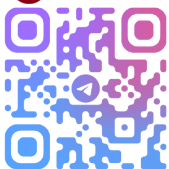
- A. 不等于 1 的实数 (A real number not equal to 1)
- B. 不等于 -1 的实数 (A real number not equal to -1)
- C. 不等于 ± 1 的实数 (A real number not equal to ± 1)
- D. 任意实数 (Any real number)

11. 已知 $z = \log_{\sin 1} \tan 1 + i \log_{\tan 1} \sin 1$, 那么 z 对应点所在象限是()

Given $z = \log_{\sin 1} \tan 1 + i \log_{\tan 1} \sin 1$, the quadrant where the corresponding point of z lies is ()

- A. 第一象限 (First quadrant)
- B. 第二象限 (Second quadrant)
- C. 第三象限 (Third quadrant)
- D. 第四象限 (Fourth quadrant)

12. 设 $z \in C$, 若 $M = z|(z - 1)^2 = |z - 1|^2$, 则 M 是()



Let $z \in \mathbb{C}$. If $M = z|(z-1)^2| = |z-1|^2$, then M is ()

- A. $M =$ 实数 ($M =$ real numbers)
- B. $M =$ 虚数 ($M =$ imaginary numbers)
- C. 实数 $\subset M \subset$ 复数 (real numbers $\subset M \subset$ complex numbers)
- D. $M =$ 复数 ($M =$ complex numbers)

13. 已知虚数 $(x-2) + yi$ ($x, y \in \mathbb{R}$) 的模为 $\sqrt{3}$, 则 $\frac{y}{x}$ 的最大值是()

Given that the modulus of the imaginary number $(x-2) + yi$ (where $x, y \in \mathbb{R}$) is $\sqrt{3}$, the maximum value of $\frac{y}{x}$ is ()

- A. $\frac{\sqrt{3}}{2}$
- B. $\frac{\sqrt{3}}{3}$
- C. $\frac{1}{2}$
- D. $\sqrt{3}$

14. 已知复数 $|z| = \frac{1}{2}$, 则 $|z^2 - z + \frac{1}{4}|$ 的最大值是()

Given that the complex number $|z| = \frac{1}{2}$, the maximum value of $|z^2 - z + \frac{1}{4}|$ is ()

- A. 2
- B. 1
- C. $\frac{3}{4}$
- D. $\frac{1}{2}$

15. 满足条件 $x|x^2 + 1| = 0, x \in \mathbb{R} \subsetneq M \subseteq m || \log_3 m + 4i| = 5, m \in \mathbb{R}^+$ 的所有集合 M 的个数是()

The number of all sets M satisfying the condition $x|x^2 + 1| = 0, x \in \mathbb{R} \subsetneq M \subseteq m || \log_3 m + 4i| = 5, m \in \mathbb{R}^+$ is ()

- A. 4
- B. 3
- C. 2
- D. 1

16. 若 z 是复数且 $|z + 5 + 12i| = 2$, 则 $|z|$ 最小值是()

If z is a complex number and $|z + 5 + 12i| = 2$, the minimum value of $|z|$ is ()

- A. 8
- B. 9
- C. 10
- D. 11

17. 若复数 z 满足 $|z + 3 - \sqrt{3}i| = \sqrt{3}$, 则 $|z|$ 的最大值为 a , $|z|$ 的最小值为 b , 则 $a + b$ 的值是()

If the complex number z satisfies $|z + 3 - \sqrt{3}i| = \sqrt{3}$, let a be the maximum value of $|z|$ and b be the minimum value of $|z|$, then the value of $a + b$ is ()

- A. $2\sqrt{3}$

B. $3\sqrt{3}$

C. $4\sqrt{3}$

D. $5\sqrt{3}$

18. 设 z 满足 $|z+i|+|z-i|=2$, 那么 $|z+i+1|$ 的最小值是()

Let z satisfy $|z+i|+|z-i|=2$, then the minimum value of $|z+i+1|$ is ()

A. 1

B. $\frac{1}{2}$

C. 2

D. 3

19. 复平面内, 曲线 $|z-1+i|=1$ 关于直线 $y=x$ 的对称曲线方程为()

In the complex plane, the equation of the symmetric curve of the curve $|z-1+i|=1$ with respect to the line $y=x$ is ()

A. $|z+1+i|=1$

B. $|\bar{z}+1+i|=1$

C. $|z-1-i|=1$

D. $|\bar{z}-1-i|=1$

20. 集合 $P = \{z \mid |z+2| + |z-2| = 6, z \in \mathbb{C}\}$, $Q = \{z \mid |z+1| = 1, z \in \mathbb{C}\}$, 则 P 与 Q 的关系是()

Let set $P = \{z \mid |z+2| + |z-2| = 6, z \in \mathbb{C}\}$ and $Q = \{z \mid |z+1| = 1, z \in \mathbb{C}\}$, then the relationship between P and Q is ()

A. $P \subset Q$

B. $Q \subset P$

C. $P = Q$

D. $P \cap Q = \emptyset$

21. 复平面内, 两点 M, N 所对应的非零复数 α, β , 满足 $\alpha^2 + \beta^2 = 0$, 则 $\triangle OMN$ (O 为坐标原点) 是()

In the complex plane, two points M and N correspond to non-zero complex numbers α and β respectively, satisfying $\alpha^2 + \beta^2 = 0$. Then $\triangle OMN$ (where O is the origin of coordinates) is ()

A. 等腰直角三角形 (Isosceles right triangle)

B. 等边三角形 (Equilateral triangle)

C. 非直角的等腰三角形 (Isosceles triangle that is not right-angled)

D. 非等腰的直角三角形 (Right triangle that is not isosceles)

22. 已知关于 x 的方程 $x^2 + x + p = 0$ 的两虚根为 x_1 和 x_2 , 且 $|x_1 - x_2| = 3$, 则实数 p 的值为()

Given that the two imaginary roots of the equation $x^2 + x + p = 0$ about x are x_1 and x_2 , and $|x_1 - x_2| = 3$, the value of the real number p is ()

A. -1

B. 1

C. $-\frac{5}{2}$



D. $\frac{5}{2}$

23. 设复数 z_1, z_2 满足 $|z_1| = |z_2| = 2$, $|z_1 + z_2| = 2\sqrt{3}$, 则 $|z_1 - z_2| = (\quad)$

Let the complex numbers z_1, z_2 satisfy $|z_1| = |z_2| = 2$ and $|z_1 + z_2| = 2\sqrt{3}$, then $|z_1 - z_2| = (\quad)$

- A. 1
- B. $\sqrt{3}$
- C. 2
- D. 3

24. 已知 i 是虚数单位, 则能使 $(n + i)^4$ 成为整数的整数 n 的个数是()

Given that i is the imaginary unit, the number of integers n that can make $(n + i)^4$ an integer is ()

- A. 2
- B. 3
- C. 4
- D. 无数个 (Countless)

25. 已知 $f(\bar{z} + i) = z + 2\bar{z} - 2i$, 则 $f(i^5)$ 等于()

Given $f(\bar{z} + i) = z + 2\bar{z} - 2i$, then $f(i^5)$ equals ()

- A. $1 + i$
- B. $1 - i$
- C. $2i$
- D. $-2i$

26. 已知 $f(n) = i^n - i^{-n}$ ($i^2 = -1$, $n \in N$), 集合 $f(n)$ 的元素个数是()

Given $f(n) = i^n - i^{-n}$ (where $i^2 = -1$, $n \in N$), the number of elements in the set $f(n)$ is ()

- A. 2
- B. 3
- C. 4
- D. 无数个 (Countless)

27. $i + i^2 + i^3 + \dots + i^{2000}$ 等于()

$i + i^2 + i^3 + \dots + i^{2000}$ equals ()

- A. 0
- B. -1
- C. 1
- D. i

28. $f(z) = 1 - \bar{z}$, $z_1 = 2 + 3i$, $z_2 = 5 - i$, 则 $\overline{f(z_1 - z_2)}$ 等于()

$f(z) = 1 - \bar{z}$, $z_1 = 2 + 3i$, $z_2 = 5 - i$, then $\overline{f(z_1 - z_2)}$ equals ()

- A. $-3 - 4i$
- B. $3 - 4i$
- C. $4 - 4i$
- D. $4 + 4i$

29. 复数 α, β 分别对应于复平面内的点 P, Q , O 为原点, 若 $\alpha^2 - 2\alpha\beta + 4\beta^2 = 0$, 则 $\triangle POQ$ 是

()

The complex numbers α, β correspond to points P, Q in the complex plane respectively, and O is the origin. If $\alpha^2 - 2\alpha\beta + 4\beta^2 = 0$, then $\triangle POQ$ is ()

- A. 等腰直角三角形 (Isosceles right triangle)
- B. 等边三角形 (Equilateral triangle)
- C. 一锐角为 60° 的直角三角形 (Right triangle with an acute angle of 60°)
- D. 顶角为 30° 的等腰三角形 (Isosceles triangle with a vertex angle of 30°)

30. 复数 $\left(\frac{1-i}{1+i}\right)^{13}$ 的值等于()

The value of the complex number $\left(\frac{1-i}{1+i}\right)^{13}$ is equal to ()

- A. $\frac{\sqrt{2}}{2}$
- B. $\sqrt{2}$
- C. i
- D. $-i$

31. 设 $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, 那么 $1 + \omega + \omega^2 + \cdots + \omega^{13}$ 的值是()

Let $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, then the value of $1 + \omega + \omega^2 + \cdots + \omega^{13}$ is ()

- A. 0
- B. $\frac{\sqrt{3}}{2} + \frac{1}{2}i$
- C. $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$
- D. $\frac{1}{2} + \frac{\sqrt{3}}{2}i$

32. 设复数 z 满足 $z + \frac{4}{z}$ 为实数, 且 $\arg(z - 2) = \frac{3\pi}{4}$, 求复数 $z = (\quad)$

Let the complex number z satisfy that $z + \frac{4}{z}$ is a real number and $\arg(z - 2) = \frac{3\pi}{4}$, then the complex number $z = (\quad)$

- A. $2i$
- B. $1 + i$
- C. $-1 + i$
- D. $1 - i$

33. 已知 z_1, z_2 是两个非 0 复数, 且 $|z_1 + z_2| = |z_1 - z_2|$, 则 $\left(\frac{z_1}{z_2}\right)^2$ 是()

Given that z_1, z_2 are two non-zero complex numbers and $|z_1 + z_2| = |z_1 - z_2|$, then $\left(\frac{z_1}{z_2}\right)^2$ is ()

- A. 正数 (Positive number)
- B. 负数 (Negative number)
- C. 零 (Zero)
- D. 虚数 (Imaginary number)